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SOLID-STATE TRANSFORMER

Abstract

A solid-state transformer is provided. The solid-state transformer includes a first stage, a first DC bus and a second stage. The first stage is configured to receive an AC power and convert the AC power into a first DC power. The first DC bus is electrically connected to the first DC power. The second stage is electrically connected to the first stage through the first DC bus, and is configured to perform isolated step-down DC/DC conversion on the first DC power to generate a second DC power. The first stage is configured to operate at a modulation index higher than 1, and the modulation index is defined by a ratio of twice AC voltage provided by the AC power to voltage across the first DC bus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application No. 63/560,494 filed on Mar. 1, 2024 and entitled "TWO-STAGE SOLID-STATE TRANSFORMER". The entire contents of the above-mentioned patent application are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION

[0002] The present disclosure relates to a solid-state transformer, and more particularly to a two-stage solid-state transformer.

BACKGROUND OF THE INVENTION

[0003] The conventional grid-connected converter usually adopts the grid interface of the line frequency transformer (LFT), which is formed by a large amount of material such as copper and iron and has a large volume. Therefore, the solid-state transformer (SST) in FIG. 1A is proposed by increasing the frequency of the transformer and adding additional solid-state devices. In this way, the volume of medium-frequency transformer (MFT) could shrink a lot. Moreover, solid-state devices provide more advanced functions such as voltage regulation, power flow control, grid-supporting, fault ride-through, and so on. As shown in FIG. 1B, SST can be used to interface MVAC (medium-voltage alternating current) grid and LVDC (low-voltage direct current) port such as renewable energy sources (e.g., photovoltaics (PV)), energy storage, electrical vehicle (EV), data center and residential load. It can also provide the MVDC (medium-voltage direct current) port for MVDC interconnection, such as railway traction and shipboard.

[0004] Up to now, the most famous and mature solid-state transformer topology for MVAC to LVDC applications is shown in FIG. 2. In order to handle high voltage, the cascade H-bridge (CHB) is adopted in MVAC side, while each isolated DC/DC converter is connected to a corresponding CHB DC port, which can be combined together and regarded as one SST module. Therefore, such a modular structure demonstrates good scalability, because the number of SST modules could be increased so that the SST can be applied to higher voltage cases.

[0005] Despite these benefits, it is still challenging to achieve high power density CHB-based SST for higher voltage classes such as 35 kV. The reason is that the AC/DC stage and isolated DC/DC stage are coupled together, where the MFT must withstand the voltage potential difference between the primary and secondary sides. When the voltage becomes higher, the larger creepage distance and insulation distance reduce the power density. A lot of space is wasted to meet high insulation requirements. Even worse, the AC grid fault such as a single-phase ground fault will cause higher phase-to-ground potential, which means the MFT insulation voltage should be higher due to the coupling between the AC/DC stage and the isolated DC/DC stage.

[0006] Therefore, there is a need of providing a solid-state transformer in order to overcome the drawbacks of the conventional technologies.

SUMMARY OF THE INVENTION

[0007] The present disclosure provides a solid-state transformer. The solid-state transformer includes a first stage and a second stage configured for performing AC/DC conversion and isolated DC/DC conversion, respectively. The first stage operates at a modulation index higher than 1 so that the insulation requirement for the second stage is reduced.

[0008] In accordance with an aspect of the present disclosure, a solid-state transformer is provided. The solid-state transformer includes a first stage, a first DC bus and a second stage. The first stage is configured to receive an AC power and convert the AC power into a first DC power. The first DC bus is electrically connected to the first DC power. The second stage is electrically connected to the first stage through the first DC bus, and is configured to perform isolated step-down DC/DC conversion on the first DC power to generate a second DC power. The first stage is configured to operate at a modulation index higher than 1, and the modulation index is defined by a ratio of twice AC voltage provided by the AC power to voltage across the first DC bus.

[0009] The above contents of the present disclosure will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1A is a schematic block diagram illustrating a conventional system including a solid-state transformer;

[0011] FIG. 1B schematically shows applications of the conventional solid-state transformer;

[0012] FIG. 2 is a schematic circuit diagram illustrating a conventional solid-state transformer;

[0013] FIG. 3 is a schematic circuit diagram illustrating a solid-state transformer according to an embodiment of the present disclosure;

[0014] FIG. 4 is a schematic circuit diagram illustrating a solid-state transformer according to another embodiment of the present disclosure;

[0015] FIG. 5 schematically shows an implementation of the first stage of the solid-state transformer in the present disclosure;

[0016] FIG. 6A and FIG. 6B schematically show two possible implementations of the submodule in FIG. 5;

[0017] FIG. 7A schematically shows simulation waveforms of the bridge arm voltage of the upper or lower bridge

arm of FIG. 5 under different modulation indexes;
 [0018] FIG. 7B schematically shows simulation waveforms of the bridge arm current flowing through the upper or lower bridge arm of FIG. 5 under different modulation indexes;
 [0019] FIG. 7C schematically shows simulation waveforms of the bridge arm energy of the upper or lower bridge arm of FIG. 5 under different modulation indexes;
 [0020] FIG. 7D schematically shows a relationship between the energy storage requirement and the modulation index at unity power factor;
 [0021] FIG. 8 schematically shows a flow chart of a zero-sequence current suppression method and corresponding architecture of a controller of the solid-state transformer according to an embodiment of the present disclosure;
 [0022] FIG. 9 schematically shows the variation of bridge arm voltage of the upper or lower bridge arm of FIG. 5 when a single-phase-ground short circuit fault occurs on the AC side of the first stage;
 [0023] FIG. 10 schematically shows a variant of the first stage of FIG. 5;
 [0024] FIG. 11 schematically show another implementation of the submodule in FIG. 5;
 [0025] FIG. 12 is a schematic circuit diagram illustrating a solid-state transformer according to another embodiment of the present disclosure;
 [0026] FIG. 13A schematically shows the second stage of FIG. 12 with a two-wire output configuration;
 [0027] FIG. 13B schematically shows the second stage of FIG. 12 with a three-wire output configuration;
 [0028] FIG. 14A and FIG. 14B schematically show two possible implementations of the DC/DC conversion module of FIG. 13A;
 [0029] FIG. 15A, FIG. 15B and FIG. 15C schematically show three possible implementations of the DC/DC conversion module of FIG. 13B;
 [0030] FIG. 16 schematically shows a variant of the second stage of FIG. 13A;
 [0031] FIG. 17A schematically shows another implementation of the second stage of the solid-state transformer of the present disclosure;
 [0032] FIG. 17B schematically shows a variant of the second stage of FIG. 17A;
 [0033] FIG. 17C is a schematic circuit diagram illustrating the switch module of the second stage of FIG. 17A and FIG. 17B;
 [0034] FIG. 18A schematically shows another implementation of the first stage of the solid-state transformer in the present disclosure;
 [0035] FIG. 18B schematically shows further another implementation of the first stage of the solid-state transformer in the present disclosure; and
 [0036] FIG. 19A and FIG. 19B exemplify two possible implementations of the switch of the upper and lower bridge arm of FIG. 18A and FIG. 18B.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0037] The present disclosure will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of preferred embodiments of this disclosure are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed.

[0038] FIG. 3 is a schematic circuit diagram illustrating a solid-state transformer according to an embodiment of the present disclosure. The solid-state transformer **100** includes a first stage **1**, a second stage **2**, and a first DC bus **3**. The first stage **1** is configured to receive an AC power from an AC source V_{ac} and convert the AC power into a first DC power. In the embodiment, the first stage **1** is configured to convert a voltage of the AC power, AC power voltage, into a voltage of the first DC power, first DC power voltage. The second stage **2** is electrically connected to the first stage **1** through the first DC bus **3** and is configured to perform isolated step-down DC/DC conversion on the first DC power to generate a second DC power, wherein the second stage **2** is adapted to convert the first DC power voltage into a voltage of the second DC power, second DC power voltage. Further, the first stage **1** is configured to operate at a modulation index M higher than **1** so that an insulation voltage required for the second stage **2** is reduced. The modulation index M is defined by a ratio of twice AC voltage provided by the AC power to voltage across the first DC bus **3**. In addition, the second stage **2** includes a DC/DC conversion module **20** with a transformer TR for isolation, and the transformer TR is for example but not limited to a medium-frequency transformer. The design of the first stage **1** and the second stage **2** could be decoupled and optimized independently. The first DC bus **3** is formed by a positive line **31** and a negative line **32**.

[0039] In an embodiment, the AC source V_{ac} provides a medium-voltage AC power, the first stage **1** includes a non-isolated MVAC/MVDC converter which converts the medium-voltage AC power into a medium-voltage DC power (first DC power), and the DC/DC conversion module **20** of the second stage **2** includes an isolated step-down MVDC/LVDC converter which converts the medium-voltage DC power into a low-voltage DC power (second DC power). It is noted that the specific voltage ranges of the foregoing high voltage, medium voltage and low voltage may be different in different technique fields and application. For example, the high voltage may be

higher than 35 kV, the medium voltage may be between 1 kV and 35 kV, and the low voltage may be lower than 1 kV.

[0040] In addition, the DC/DC conversion module **20** of the second stage **2** further includes a DC/AC conversion circuit **21** and an AC/DC conversion circuit **22**. The transformer TR is coupled between the DC/AC conversion circuit **21** and the AC/DC conversion circuit **22**. The output of the AC/DC conversion circuit **22** may be unipolar or bipolar and is connected to a positive terminal Vo+ and a negative terminal Vo- of a second DC bus. For example, the second DC bus may be connected to a low-voltage DC distribution grid, a distributed energy source, an energy storage system, or other flexible loads. Additionally, the first DC bus **3** may be connected to a medium-voltage DC distribution grid, but not limited thereto.

[0041] In an embodiment, the solid-state transformer **100** includes two capacitors C1 and C2 electrically connected in series across the first DC bus **3** and form a midpoint O therebetween. The midpoint O is grounded with an impedance Z1. Accordingly, an insulation voltage required for the transformer TR of the second stage **2** could be reduced to a half of the voltage across the first DC bus **3**. In this embodiment, the midpoint O is further electrically connected to the DC/AC conversion circuit **21**. In addition, the impedance between the AC source Vac and the ground terminal is represented by an impedance Z2. In an embodiment, a port of the output of the AC/DC conversion circuit **22** may be grounded with an impedance Z3.

[0042] Various grounding schemes can be adopted between the midpoint O and the ground terminal, such as solid grounding, low-impedance grounding, high-impedance grounding, resonant grounding or direct grounding. In an embodiment, the AC source Vac is provided by a substation, the grounding scheme adopted between the midpoint O and the ground terminal depends on the grounding scheme of the substation. For different areas around the world, the grounding scheme of the substation and the corresponding grounding scheme of the midpoint O may be different. For instance, when the substation provides MVAC and the first and second stages **1** and **2** perform MVAC/MVDC conversion and MVDC/LVDC conversion respectively, in North America, the grounding scheme of the substation may adopt solid-grounding or low-impedance grounding. Correspondingly, the midpoint O is ungrounded, or the grounding scheme of the midpoint O adopts high-impedance grounding. In China and Europe, the grounding scheme of the substation may adopt resonant grounding, or the substation may be ungrounded. Correspondingly, MVAC side adopts grounding transformer, and the midpoint O is ungrounded.

[0043] In addition, when the solid-state transformer **100** works in the islanding mode and is disconnected with the substation, the grounding scheme may be applied as the system functional grounding. For example, a high-voltage switch and a series low impedance grounding resistor may be connected to the midpoint O. Under the islanding mode, the high-voltage switch is closed, and the midpoint O becomes the new system functional grounding point.

[0044] FIG. 4 is a schematic circuit diagram illustrating a solid-state transformer according to another embodiment of the present disclosure. The component parts and elements corresponding to those of FIG. 3 are designated by identical numeral references, and detailed descriptions thereof are omitted herein. In the solid-state transformer **100** of this embodiment, the second stage **2** may include multiple DC/DC conversion modules **20** (exemplified by two DC/DC conversion modules **20** in the figure). The input of each DC/DC conversion module **20** is electrically connected to the first DC bus **3**, and the outputs of the multiple DC/DC conversion modules **20** may be electrically connected to different DC systems through different second DC buses. In other words, in this embodiment, the solid-state transformer **100** can interface different DC systems. In another embodiment, the outputs of the multiple DC/DC conversion modules **20** may be electrically connected to the same DC system through the same second DC bus. In an embodiment, the solid-state transformer **100** further includes switches SW1 and SW2 electrically connected between the first DC bus **3** and each DC/DC conversion module **20** of the second stage **2**. The switches SW1 and SW2 may be controlled to enable or disable the DC/DC conversion module **20**. In an aspect, the switches SW1 and SW2 may be used to isolate different DC/DC conversion modules **20**. For example, when the short circuit failure happens at the output of one DC/DC conversion module **20**, the corresponding switches SW1 and SW2 could be triggered to turn off for isolating this DC/DC conversion module **20**, while the other DC/DC conversion modules **20** can still operate properly without influence. In addition, in this embodiment, the impedance between the primary side of the DC/DC conversion module **20** and the ground terminal is represented by the impedance Z11. It is noted that the impedances Z11 or Z3 corresponding to different DC/DC conversion modules **20** may be different.

[0045] FIG. 5 schematically shows an implementation of the first stage of the solid-state transformer in the present disclosure. As shown in FIG. 5, in an embodiment, the first stage **1** includes a modular multilevel converter. In specific, the first stage **1** includes three upper bridge arms and three lower bridge arms. The three upper bridge arms are identical, and each upper bridge arm includes an inductor L and submodules SM electrically connected in series between the positive line **31** of the first DC bus **3** and the corresponding phase of the AC source Vac. The three lower bridge arms are identical, and each lower bridge arm includes an inductor L and submodules SM electrically connected in series between the negative line **32** of the first DC bus **3** and the corresponding phase of the AC source Vac.

[0046] The submodule SM may be implemented by any suitable topology. FIG. 6A and FIG. 6B schematically show two possible implementations of the submodule in FIG. 5. In an embodiment, as shown in FIG. 6A, the submodule SM may be unipolar and includes a half-bridge circuit including switches S1 and S2 and a capacitor Csm. The switches S1 and S2 are serially connected to form a switch bridge electrically connected in parallel to the capacitor Csm. The connection node of the switches S1 and S2 is coupled to a port P1, and a negative terminal of the capacitor Csm is coupled to a port P2. In another embodiment, as shown in FIG. 6B, the submodule SM may be bipolar and includes a full-bridge circuit including switches S1, S2, S3, and S4 and a capacitor Csm. The bipolar submodule SM can generate negative arm voltage and block fault. The switches S1 and S2 are serially connected to form a first bridge arm, the switches S3 and S4 are serially connected to form a second bridge arm, and the first and second bridge arms and the capacitor Csm are electrically connected in parallel. The connection node of the switches S1 and S2 are coupled to a port P1, and the connection node of the switches S3 and S4 are coupled to a port P2. Refer to FIGS. 6A and 6B in conjunction with FIG. 5. Regarding the submodule SM of the upper bridge arm in FIG. 5, the port P1 of the submodule SM is coupled to the port P2 of the neighboring submodule SM or the positive line 31, and the port P2 of the submodule SM is coupled to the port P1 of the neighboring submodule SM or the inductor L. Regarding the submodule SM of the lower bridge arm in FIG. 5, the port P1 of the submodule SM is coupled to the port P2 of the neighboring submodule SM or the inductor L, and the port P2 of the submodule SM is coupled to the port P1 of the neighboring submodule SM or the negative line 32.

[0047] The capacitor Csm of each submodule SM may be controlled to stabilize a voltage Vsm across the capacitor Csm. In the embodiment, the modulation index M of the first stage 1 is defined in equation (1).

$$[00001] \quad M = \frac{2V_{ac}}{V_{dc}} \quad (1)$$

[0048] In equation (1), Vac represents the peak value of the AC voltage provided by the AC source Vac, and Vdc represents the voltage across the first DC bus 3. In the embodiment, the modulation index is the ratio of twice AC voltage provided by the AC power to voltage across the first DC bus 3.

[0049] Neglecting the voltage across the inductor L, the bridge arm voltage Varm of each upper or lower bridge arm may be calculated as equation (2), where ω represents the angular frequency.

$$[00002] \quad V_{arm} = 0.5V_{dc} - V_{ac} \cdot \text{Math. sin}(\omega t) = \frac{V_{ac}}{4M} - V_{ac} \cdot \text{Math. sin}(t) \quad (2)$$

[0050] The simulation waveforms of the bridge arm voltage Varm under different modulation indexes M are plotted in FIG. 7A. In FIG. 7A, waveforms 71, 72, and 73 represent the bridge arm voltage Varm under modulation indexes M equal to 0.8, 1 and 1.4, respectively. As shown in FIG. 7A, the bridge arm voltage Varm appears negative value when the modulation index M is higher than 1.

[0051] The number N of submodules SM in each upper or lower bridge arm is determined by the maximum bridge arm voltage stress.

$$[00003] \quad N = \text{Math. } \frac{\max(V_{arm})}{V_{sm}} \cdot \text{Math.} \quad (3)$$

[0052] According to equation (3), a higher modulation index M indicates the lower peak of bridge arm voltage Varm as well as the lower number N of submodules SM.

[0053] The bridge arm current Iarm flowing through each upper or lower bridge arm could be calculated as:

$$[00004] \quad I_{arm} = \frac{I_{dc}}{3} + 0.5I_{ac} \cdot \text{Math. sin}(t) = \frac{M}{4}I_{ac} + 0.5I_{ac} \cdot \text{Math. sin}(t) \quad (4)$$

[0054] In equation (4), Idc is the current flowing through the first DC bus 3, and Iac is the current provided by the AC source Vac. According to equation (4), when the modulation index M increases, the root-mean-square value of the bridge arm current Iarm becomes higher. The simulation waveforms of the bridge arm current Iarm under different modulation indexes M are plotted in FIG. 7B. In FIG. 7B, waveforms 74, 75 and 76 represent the bridge arm current Iarm under modulation indexes M equal to 0.8, 1 and 1.4, respectively.

[0055] Based on equations (2) and (4), the bridge arm power Parm of each upper or lower bridge arm could be calculated as:

[00005]

$$P_{arm} = V_{arm} \cdot \text{Math. } I_{arm} = \frac{I_{dc} \cdot \text{Math. } V_{dc}}{2} - \frac{V_{ac} \cdot \text{Math. } I_{ac}}{4} + \left(\frac{1}{2M} - \frac{M}{4}\right)I_{ac} \cdot \text{Math. } V_{ac} \cdot \text{Math. sin}(t) + \frac{V_{ac} \cdot \text{Math. } I_{ac}}{4}\cos(2t) \quad (5)$$

[0056] In equation (5), the DC term would be cancelled naturally, while the amplitude of the line-frequency term is greatly impacted by the modulation index M. When M=1.4, the coefficient substantially becomes zero. Therefore, the bridge arm energy Earm, which equals the integral of bridge arm power Parm (equation (6)), has the minimum ripple. The simulation waveforms of the bridge arm energy Earm under different modulation indexes M are plotted in FIG. 7C. In FIG. 7C, waveforms 80, 81 and 82 represent the bridge arm energy Earm under modulation indexes M equal to 0.8, 1 and 1.4, respectively.

$$[00006] \quad E_{arm} = \int P_{arm} dt \quad (6)$$

[0057] Since the capacitor Csm of submodule SM confronts the line frequency energy ripple and should be large enough to suppress the voltage ripple, the capacitor Csm greatly impacts the power density of submodule SM. Therefore, the energy storage requirement Eunit is usually applied and could be calculated as:

$$[00007] \text{ Eunit} = \frac{3}{2} \frac{\text{Earm}}{S} = \frac{3(\text{Earm, max} - \text{Earm, min})}{2S} \quad (7)$$

[0058] In equation (7), $\Delta E_{\text{sub.arm}}$ denotes the energy ripple across each upper or lower bridge arm and can be calculated as the difference between the maximum value Earm,max and the minimum Earm,min of the bridge arm energy Earm . Moreover, ϵ refers to the desired voltage ripple coefficient, and S means the apparent power of the whole system.

[0059] Using the ideal waveform, the relationship between the energy storage requirement Eunit and the modulation index M at unity power factor are plotted in FIG. 7D. It can be observed that when the modulation index M is around 1.41, the energy storage requirement Eunit reaches 6.71 kJ/MVA, which is only 39% of that with $M=1$. Therefore, the high modulation index M helps to reduce the energy storage requirement Eunit . Moreover, the higher modulation index M leads to the lower voltage V_{dc} across the first DC bus 3, which also reduces the insulation voltage required for the transformer TR in the second stage 2.

[0060] Refer to FIG. 3 again. The impedance of the zero-sequence loop mainly depends on the impedance Z_2 , and a zero-sequence current may flow through the midpoint O and a neutral point of the AC source V_{ac} when the impedances Z_2 and Z_1 are low. FIG. 8 schematically shows a flow chart of a zero-sequence current suppression method and corresponding architecture of a controller of the solid-state transformer according to an embodiment of the present disclosure. As shown in FIG. 8, the three-phase currents $i_{\text{sub.a.b.c}}$ of the AC source V_{ac} are measured and fed into two different paths to achieve three-phase reference voltages $v_{\text{sub.a.b.c}}^*$. In specific, through the upper path, firstly, the three-phase currents $i_{\text{sub.a.b.c}}$ are transformed into a d-axis current I_d and a q-axis current I_q by a converter 41. Then, the d-axis current I_d and the q-axis current I_q are compared with their respective reference values I_d^* and I_q^* to generate error signals by comparators 42 and 43. Afterwards, the error signals from the comparators 42 and 43 are respectively processed by two proportional-integral (PI) controllers 44 and 45 to generate control signals for the d-axis and q-axis. Finally, the control signals from the PI controllers 44 and 45 are transformed back into three-phase voltage signals by a converter 46. In addition, through the lower path, firstly, the three-phase currents $i_{\text{sub.a.b.c}}$ are summed up to generate a zero-sequence current I_O by a calculator 47. Then, the zero-sequence current I_O is compared with zero to generate a difference by a comparator 48. Afterwards, the difference from the comparator 48 is processed by a regulator 49 to generate a regulation value utilized for suppressing the zero-sequence current, and the regulator 49 may be implemented by a proportional resonant or repetitive control regulator. The outputs of the two paths, namely the three-phase voltage signals from the converter 46 and the regulation value from the regulator 49, are summed up by an adder 50 to generate the three-phase reference voltages $v_{\text{sub.a.b.c}}^*$. It should be noted that the converters, comparators, PI controllers, calculator, regulator, and adder shown FIG. 8 may be implemented as hardware or software modules. If implemented as hardware, they can be realized by separate hardware components or integrated into a smaller number of hardware units.

[0061] FIG. 9 schematically shows the variation of bridge arm voltage V_{arm} when a single-phase-ground (SPG) short circuit fault occurs on the AC side of the first stage 1 in case of low impedance Z_1 . In FIG. 9, the waveform of bridge arm voltage V_{arm} is depicted by solid lines, the magnitude of $0.5 \cdot V_{\text{dc}}$ is depicted by dashed lines, T_1 represents the period under normal operation, and T_2 represents the period during the SPG short circuit fault. When a SPG short circuit fault occurs in one phase, the voltage amplitude of the other two phases would increase, and the increment depends on the impedance Z_2 (i.e., the grounding impedance of the AC source V_{ac} as shown in FIG. 3). According to equation (2), the voltage V_{dc} across the first DC bus 3 decreases correspondingly so that the peak value of the bridge arm voltage V_{arm} under SPG short circuit fault does not exceed that under the normal operation. During the SPG short circuit fault, the number of submodules SM of each upper or lower bridge arm should be sufficient to handle the negative overvoltage of AC power. Then, the grounding current could be controlled to be zero, which means the first stage 1 rides through the SPG short circuit fault. As for the second stage 2, since the voltage V_{dc} across the first DC bus 3 reduces, the transformer TR in the second stage 2 doesn't have an overvoltage concern that exists in the conventional approaches.

[0062] Taking the implementation of the first stage 1 in FIG. 5 as an example, the first stage 1 further includes three varistors 11, 12, and 13 applied to the input of first stage 1 for lightning protection, as exemplified in FIG. 10. Accordingly, the overvoltage from the lightning could be clamped. The three varistors 11, 12, and 13 may be connected in Δ -connection as shown in embodiment of FIG. 10, or may be connected in Y-connection in another embodiment. The varistors 11, 12, and 13 are for example but not limited to metal-oxide varistors. Furthermore, taking the full-bridge submodules SM in FIG. 6B as an example, a damping resistor R_d and a bypass switch S_b may be applied in each submodule SM to avoid the overcharging of the capacitor, as exemplified in FIG. 11. As shown in FIG. 11, the damping resistor R_d and the bypass switch S_b are electrically connected between the ports P1 and P2 of the submodule SM. For instance, the bypass switch S_b may include a bidirectional semiconductor device (e.g., IGBT), a thyristor, or a mechanical switch.

[0063] FIG. 12 is a schematic circuit diagram illustrating a solid-state transformer according to another

embodiment of the present disclosure. The component parts and elements corresponding to those of FIG. 3 and FIG. 5 are designated by identical numeral references, and detailed descriptions thereof are omitted herein. In the solid-state transformer **100** of this embodiment, the second stage **2** includes a plurality of DC/DC conversion modules **20**, and the plurality of DC/DC conversion modules **20** are divided into first and second DC/DC conversion modules **20a** and **20b** according to the disposing position. In specific, the second stage **2** includes a plurality of first DC/DC conversion modules **20a** and a plurality of second DC/DC conversion modules **20b**. The inputs of the plurality of first DC/DC conversion modules **20a** are electrically connected in series between the positive line **31** of the first DC bus **3** and the midpoint O, the inputs of the plurality of second DC/DC conversion modules **20b** are electrically connected in series between the midpoint O and the negative line **32** of the first DC bus **3**, and the outputs of all the first and second DC/DC conversion modules **20a** and **20b** are electrically connected in parallel and coupled to the positive and negative terminals Vo+ and Vo- of the second DC bus. The ISOP (input-series-output-parallel) configuration in this embodiment could be used to handle the voltage Vdc across the first DC bus **3** and the large current in the second DC bus. In an embodiment, the first and second DC/DC conversion modules **20a** and **20b** have the same number, and the total number thereof is determined by the voltage Vdc across the first DC bus **3** and is calculated as:

$$[00008] X = \frac{V_{dc}}{V_{dc_module}} \quad (8)$$

[0064] In equation (8), X represents the total number of the first and second DC/DC conversion modules **20a** and **20b**, and Vdc_module represents the voltage handled by each of the first and second DC/DC conversion modules **20a** and **20b**.

[0065] In an embodiment, the input of each of the first and second DC/DC conversion modules **20a** and **20b** may be electrically connected in parallel to an input capacitor Cin, and the output of each of the first and second DC/DC conversion modules **20a** and **20b** may be electrically connected in parallel to an output capacitor Co.

[0066] Depending on the port requirement of the DC system connected to the second DC bus, there may have different configurations for the output of the second stage **2**.

[0067] FIG. 13A schematically shows the second stage of FIG. 12 with a two-wire output configuration. As shown in FIG. 13A, the output of the AC/DC conversion circuit **22** of each of the first and second DC/DC conversion modules **20a** and **20b** adopts two-wire configuration. In other words, the output of each AC/DC conversion circuit **22** has two terminals electrically connected to the positive and negative terminals Vo+ and Vo- of the second DC bus respectively. Under this circumstance, each DC/DC conversion module may provide a unipolar output or a bipolar output.

[0068] FIG. 13B schematically shows the second stage of FIG. 12 with a three-wire output configuration. As shown in FIG. 13B, the output of the AC/DC conversion circuit **22** of each of the first and second DC/DC conversion modules **20a** and **20b** adopts three-wire configuration. In other words, the output of each AC/DC conversion circuit **22** has three terminals electrically connected to the ground terminal and the positive and negative terminals Vo+ and Vo- of the second DC bus respectively. Under this circumstance, each DC/DC conversion module may provide a bipolar output.

[0069] In addition, the redundancy design is possible for such kind of ISOP structure to improve reliability. Therefore, in an embodiment, a bypass circuit may be electrically connected in parallel to the input of each DC/DC conversion module. Refer to FIG. 13A again. Taking the bypass circuit corresponding to the top one of the first DC/DC conversion modules **20a** in FIG. 13A as an example, the bypass circuit includes a switch Q1, a damping resistor Rd2 and a switch Q2. The switch Q1 and the damping resistor Rd2 are electrically connected in series to form a branch, which is electrically connected in parallel to the switch Q2 and the input of the first DC/DC conversion module **20a**. Preferably, the switch Q1 is a semiconductor switch, and the switch Q2 is a mechanical switch, but not exclusively. Consequently, the solid-state transformer **100** can operate with fault DC/DC conversion modules in the second stage **2**. For example, when faults occur in a DC/DC conversion module, the fault DC/DC conversion module can be bypassed by turning on the switches Q1 and Q2. Therefore, the reliability of the solid-state transformer **100** is improved.

[0070] For the second stage **2** with two-wire output configuration, two possible implementations of the DC/DC conversion module (**20a** or **20b**) of FIG. 13A are exemplified in FIG. 14A and FIG. 14B.

[0071] In the implementation shown in FIG. 14A, each of the first and second DC/DC conversion modules **20a** and **20b** includes a two-level CLLC resonant converter. The DC/AC conversion circuit **21** includes two primary switch bridge arms electrically connected in parallel, and each primary switch bridge arm includes two serially-connected switches. The DC/AC conversion circuit **21** further includes a resonant capacitor Cr1 and a resonant inductor Lr1 electrically connected in series between a connection node of the switches of one primary switch bridge arm and a first terminal of a primary winding of the transformer TR. The second terminal of the primary winding of the transformer TR is coupled to a connection node of the switches of the other primary switch bridge arm. The AC/DC conversion circuit **22** includes two secondary switch bridge arms electrically connected in parallel, and each

secondary switch bridge arm includes two serially-connected switches. The AC/DC conversion circuit **22** further includes a resonant capacitor Cr2 and a resonant inductor Lr2 electrically connected in series between a connection node of the switches of one secondary switch bridge arm and a first terminal of a secondary winding of the transformer TR. The second terminal of the secondary winding of the transformer TR is coupled to a connection node of the switches of the other secondary switch bridge arm. Two output terminals of the AC/DC conversion circuit **22** are electrically connected to the positive and negative terminals Vo+ and Vo- of the second DC bus, respectively.

[0072] In the implementation shown in FIG. **14B**, each of the first and second DC/DC conversion modules **20a** and **20b** includes a dual-active-bridge converter. The components and connection relations are similar with that of FIG. **14A** except that the first terminal of the primary winding of the transformer TR is coupled to the connection node of the primary switch bridge arm of the DC/AC conversion circuit **21** through an inductor Lk, and the first terminal of the secondary winding of the transformer TR is coupled to the connection node of secondary switch bridge arm of the AC/DC conversion circuit **22**. Two output terminals of the AC/DC conversion circuit **22** are electrically connected to the positive and negative terminals Vo+ and Vo- of the second DC bus, respectively

[0073] It is noted that the possible topology of the DC/DC conversion module of FIG. **13A** is not limited to the implementations shown in FIG. **14A** and FIG. **14B**. For example, the DC/DC conversion module of FIG. **13A** may adopt any suitable three-level or three-phase circuit topology.

[0074] For the second stage **2** with three-wire output configuration, three possible implementations of the DC/DC conversion module (**20a** or **20b**) of FIG. **13B** are exemplified in FIG. **15A**, FIG. **15B** and FIG. **15C**.

[0075] In the implementation shown in FIG. **15A**, the transformer TR may be a three-winding transformer used to interface three full-bridges. The structure of DC/AC conversion circuit **21** is similar to that in FIG. **14A**. The AC/DC conversion circuit **22** includes two full-bridge circuits connected to two secondary windings of the transformer TR, and each full-bridge circuit has the structure similar with the secondary circuit in FIG. **14A**. Two output terminals of one full-bridge circuit of the AC/DC conversion circuit **22** are electrically connected to the positive terminals Vo+ of the second DC bus and the ground terminal, respectively. Two output terminals of the other full-bridge circuit of the AC/DC conversion circuit **22** are electrically connected to the ground terminal and the negative terminal Vo- of the second DC bus, respectively. Accordingly, in this implementation, the second stage **2** provides a bipolar output.

[0076] In the implementation shown in FIG. **15B**, the structure of DC/AC conversion circuit **21** is similar to that in FIG. **15A**, and the AC/DC conversion circuit **22** adopts a series H-bridge structure. In particular, the AC/DC conversion circuit **22** includes four serial-connected switches, a resonant inductor Lr2 and a resonant capacitor Cr2. The connection node of the first and second switches is electrically connected to a first terminal of a secondary winding of the transformer TR through the resonant inductor Lr2 and the resonant capacitor Cr2, and the connection node of the third and fourth switches is electrically connected to a second terminal of the secondary winding of the transformer TR. The AC/DC conversion circuit **22** further includes one capacitor electrically connected in parallel to the first and second switches and another capacitor electrically connected in parallel to the third and fourth switches. The first switch is electrically connected to the positive terminal Vo+ of the second DC bus, the connection node of the second and third switches is electrically connected to the ground terminal, and the fourth switch is electrically connected to the negative terminal Vo- of the second DC bus. Accordingly, in this implementation, the second stage **2** provides a bipolar output.

[0077] In the implementation shown in FIG. **15C**, the DC/AC conversion circuit **21** includes a first primary circuit **21a** and a second primary circuit **21b**, the transformer TR includes a first transformer TR1 and a second transformer TR2, and the AC/DC conversion circuit **22** includes a first secondary circuit **22a** and a second secondary circuit **22b**. The first transformer TR1 is coupled between the first primary circuit **21a** and the first secondary circuit **22a**, and the second transformer TR2 is coupled between the second primary circuit **21b** and the second secondary circuit **22b**. Each of the first and second primary circuits **21a** and **21b** has the structure similar to that of the DC/AC conversion circuit **21** in FIG. **14A**, and the inputs of the first and second primary circuits **21a** and **21b** are electrically connected in parallel. Each of the first and second secondary circuits **22a** and **22b** has the structure similar to that of the AC/DC conversion circuit **22** in FIG. **14B**. Two output terminals of the first secondary circuit **22a** are electrically connected to the positive terminal Vo+ of the second DC bus and the ground terminal respectively. Two output terminals of the second secondary circuit **22b** are electrically connected to the ground terminal and the negative terminal Vo- of the second DC bus respectively. Accordingly, in this implementation, the second stage **2** provides a bipolar output.

[0078] In addition, in the above embodiments, all the first and second DC/DC conversion modules **20a** and **20b** are electrically connected to the same second DC bus, namely the second stage **2** provides one output for one DC system. However, the present disclosure is not limited thereto. In another embodiment, the DC/DC conversion modules may provide different outputs to different DC systems (DC loads) through different second DC buses. For example, in an embodiment, as shown in FIG. **16**, the outputs of the plurality of first DC/DC conversion modules

20a may be electrically connected in parallel and coupled to a positive terminal Vol+ and a negative terminal Vol- of one second DC bus, and the outputs of the plurality of second DC/DC conversion modules **20b** may be electrically connected in parallel and coupled to a positive terminal Vo2+ and a negative terminal Vo2- of another second DC bus.

[0079] FIG. **17A** schematically shows another implementation of the second stage of the solid-state transformer of the present disclosure. In this implementation, as shown in FIG. **17A**, the second stage **2** includes a modular multilevel resonant converter. Specifically, the second stage **2** includes a plurality of switch modules SS and a plurality of conversion units **23** (exemplified by two conversion units **23** in the figure). Half of the switch modules SS are electrically connected in series between the positive line **31** of the first DC bus **3** and a midpoint O2, and the other half of the switch modules SS are electrically connected in series between the midpoint O2 and the negative line **32** of the first DC bus **3**. Each conversion unit **23** has a first input terminal **231**, a second input terminal **232**, a first output terminal **233** and a second output terminal **234**, and includes a resonant tank, a transformer TR3 and a full-bridge circuit. In specific, in each conversion unit **23**, a resonant inductor Lr and a resonant capacitor Cr of the resonant tank are electrically connected in series between the first input terminal **231** and a first terminal of a primary winding of the transformer TR3, and a second terminal of the primary winding of the transformer TR3 is electrically connected to the second input terminal **232**. The full-bridge circuit includes two parallel-connected bridge arms, where one bridge arm includes switches Q3 and Q4 connected in series, and the other bridge arm includes switches Q5 and Q6 connected in series. Two terminals of a secondary winding of the transformer TR3 are respectively connected to a connection node of the switches Q3 and Q4 and a connection node of the switches Q5 and Q6.

[0080] The inputs of the conversion units **23** may be electrically connected in series or parallel. For example, in the embodiment shown in FIG. **17A**, the inputs of the conversion units **23** are electrically connected in parallel. In specific, the first input terminals **231** of the conversion units **23** are all connected to the midpoint O2, and the second input terminals **232** of the conversion units **23** are all connected to the midpoint O. In another embodiment, as shown in FIG. **17B**, the inputs of the conversion units **23** are electrically connected in series between the midpoint O and the midpoint O2. In specific, the first input terminal **231** of the first one of the conversion units **23** is connected to the midpoint O2, and the second input terminal **232** of the last one of the conversion units **23** is connected to the midpoint O. Among two neighboring conversion units **23**, the first input terminal **231** of one conversion unit **23** is connected to the second input terminal **232** of the other conversion unit **23**. Depending on the DC systems powered by the solid-state transformer **100**, the first and second output terminals **233** and **234** of different conversion units **23** may be electrically connected to same second DC bus or different second DC buses.

[0081] FIG. **17C** is a schematic circuit diagram illustrating the switch module SS of the second stage **2** of FIG. **17A** and FIG. **17B**. In an embodiment, as shown in FIG. **17C**, each switch module SS has ports P3 and P4 and includes a capacitor C_{ss} and switches Q_{ss1} and Q_{ss2}. A first terminal of the switch Q_{ss1} is electrically connected to the port P3 and a first terminal of the switch Q_{ss2}, a second terminal of the switch Q_{ss1} is electrically connected to a positive terminal of the capacitor C_{ss}, and a negative terminal of the capacitor C_{ss} is electrically connected to a second terminal of the switch Q_{ss2} and the port P4. The capacitor C_{ss} has a rated voltage V_{ss}. Please refer to FIG. **17C** in conjunction with FIG. **17A**. The number of the switch modules SS could be determined by dividing the voltage V_{dc} across the first DC bus **3** by the rated voltage V_{ss}.

[0082] FIG. **18A** schematically shows another implementation of the first stage of the solid-state transformer in the present disclosure. The component parts and elements corresponding to those of FIG. **5** are designated by identical numeral references, and detailed descriptions thereof are omitted herein. In FIG. **18A**, the three upper bridge arms, from left to right, are referred as the first, second and third upper bridge arms, respectively, and the three lower bridge arms, from left to right, are referred as the first, second and third lower bridge arms, respectively. Different from the embodiment of FIG. **5**, in this embodiment, each of the three upper bridge arms and the three lower bridge arms further includes a switch connected between the inductor L and the corresponding phase of the AC source Vac. Specifically, the inductors L of the first upper and lower bridge arms are electrically connected to the first phase of the AC source Vac through switches S11 and S14 respectively, the inductors L of the second upper and lower bridge arms are electrically connected to the second phase of the AC source Vac through switches S12 and S15 respectively, and the inductors L of the third upper and lower bridge arms are electrically connected to the third phase of the AC source Vac through switches S13 and S16 respectively.

[0083] FIG. **18B** schematically shows further another implementation of the first stage of the solid-state transformer in the present disclosure. The component parts and elements corresponding to those of FIG. **5** are designated by identical numeral references, and detailed descriptions thereof are omitted herein. Different from the embodiment of FIG. **5**, in this embodiment, each of the three upper bridge arms and the three lower bridge arms further includes two switches, where one switch is connected between the inductor L and the corresponding phase of the AC source Vac, and the other switch is connected between the inductor and the midpoint O. In specific, the first upper bridge arm includes a switch S11a, electrically connected between the inductor L and the first phase of

the AC source Vac, and a switch S11b, electrically connected between the inductor L and the midpoint O. The second upper bridge arm includes a switch S12a, electrically connected between the inductor L and the second phase of the AC source Vac, and a switch S12b, electrically connected between the inductor L and the midpoint O. The third upper bridge arm includes a switch S13a, electrically connected between the inductor L and the third phase of the AC source Vac, and a switch S13b, electrically connected between the inductor L and the midpoint O. The first lower bridge arm includes a switch S14a, electrically connected between the inductor L and the first phase of the AC source Vac, and a switch S14b, electrically connected between the inductor L and the midpoint O. The second lower bridge arm includes a switch S15a, electrically connected between the inductor L and the second phase of the AC source Vac, and a switch S15b, electrically connected between the inductor L and the midpoint O. The third lower bridge arm includes a switch S16a, electrically connected between the inductor L and the third phase of the AC source Vac, and a switch S16b, electrically connected between the inductor L and the midpoint O. In a word, in each upper or lower bridge arm, the inductor L is electrically connected to the corresponding phase of the AC source Vac through a switch and is electrically connected to the midpoint O through another switch.

[0084] In the present embodiment, each of the foregoing switches of the upper and lower bridge arms shown in FIG. 18A and FIG. 18B may be formed by series-connected insulated gate bipolar transistors shown in FIG. 19A or by series-connected metal-oxide-semiconductor field-effect transistors shown in FIG. 19B.

[0085] While the disclosure has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the disclosure needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A solid-state transformer, comprising: a first stage, configured to receive an AC power and convert the AC power into a first DC power; a first DC bus, electrically connected to the first DC power; and a second stage, electrically connected to the first stage through the first DC bus, and configured to perform isolated step-down DC/DC conversion on the first DC power to generate a second DC power, wherein the first stage is configured to operate at a modulation index higher than 1, wherein the modulation index is defined by a ratio of twice AC voltage provided by the AC power to voltage across the first DC bus.
2. The solid-state transformer according to claim 1, further comprising two capacitors electrically connected in series across the first DC bus and a midpoint between the two capacitors.
3. The solid-state transformer according to claim 2, wherein the midpoint is grounded by solid grounding, low-impedance grounding, high-impedance grounding, resonant grounding or direct grounding.
4. The solid-state transformer according to claim 1, wherein the second DC power is unipolar.
5. The solid-state transformer according to claim 1, wherein the second DC power is bipolar.
6. The solid-state transformer according to claim 1, wherein the second stage comprises DC/DC conversion modules electrically connected to the first DC bus, and inputs of the DC/DC conversion modules are electrically connected in parallel.
7. The solid-state transformer according to claim 6, wherein outputs of the DC/DC conversion modules are electrically connected to different second DC buses.
8. The solid-state transformer according to claim 6, wherein outputs of the DC/DC conversion modules are electrically connected in parallel and are connected to a second DC bus.
9. The solid-state transformer according to claim 6, further comprising a switch electrically connected between the first DC bus and each of the DC/DC conversion modules, wherein the switch is configured to be controlled to enable or disable a corresponding one of the DC/DC conversion modules.
10. The solid-state transformer according to claim 1, wherein the second stage comprises first DC/DC conversion modules and second DC/DC conversion modules, inputs of the first DC/DC conversion modules are electrically connected in series between a positive line of the first DC bus and a midpoint, and inputs of the second DC/DC conversion modules are electrically connected in series between a negative line of the first DC bus and the midpoint.
11. The solid-state transformer according to claim 10, wherein an output of each of the first DC/DC conversion modules and the second DC/DC conversion modules is unipolar or bipolar, and has two terminals electrically connected to a positive terminal and a negative terminal of a second DC bus, respectively.
12. The solid-state transformer according to claim 10, wherein an output of each of the first DC/DC conversion modules and the second DC/DC conversion modules is bipolar, and has three terminals electrically connected to a positive terminal and a negative terminal of a second DC bus and a ground terminal, respectively.
13. The solid-state transformer according to claim 10, wherein outputs of the first DC/DC conversion modules and

the second DC/DC conversion modules are electrically connected in parallel and are connected to a second DC bus.

14. The solid-state transformer according to claim 10, wherein outputs of the first DC/DC conversion modules are electrically connected in parallel and are connected to a second DC bus, and outputs of the second DC/DC conversion modules are electrically connected in parallel and are connected to a third DC bus.

15. The solid-state transformer according to claim 10, further comprising a bypass circuit electrically connected in parallel to each of the inputs of the first DC/DC conversion modules and the second DC/DC conversion modules, wherein the bypass circuit comprises a first switch, a damping resistor and a second switch, and the second switch is electrically connected in parallel to a branch formed by the first switch and damping resistor connected in series.

16. The solid-state transformer according to claim 1, further comprising two capacitors electrically connected in series across the first DC bus and a first midpoint between the two capacitors, wherein the second stage comprises first switch modules, second switch modules, a second midpoint and conversion units, the first switch modules are electrically connected in series between a positive line of the first DC bus and the second midpoint, the second switch modules are electrically connected in series between the second midpoint and a negative line of the first DC bus, inputs of the conversion units are electrically connected in parallel, and each of the inputs has two terminals coupled to the first midpoint and the second midpoint respectively.

17. The solid-state transformer according to claim 1, further comprising two capacitors electrically connected in series across the first DC bus and a first midpoint between the two capacitors, wherein the second stage comprises first switch modules, second switch modules, a second midpoint and conversion units, the first switch modules are electrically connected in series between a positive line of the first DC bus and the second midpoint, the second switch modules are electrically connected in series between the second midpoint and a negative line of the first DC bus, and inputs of the conversion units are electrically connected in series between the first midpoint and the second midpoint.

18. The solid-state transformer according to claim 1, wherein the first stage comprises three upper bridge arms and three lower bridge arms, each of the three upper bridge arms comprises an inductor and submodules electrically connected in series between a positive line of the first DC bus and a corresponding phase of the AC power, and each of the three lower bridge arms comprises an inductor and submodules electrically connected in series between a negative line of the first DC bus and a corresponding phase of the AC power.

19. The solid-state transformer according to claim 18, wherein each of the submodules of each of the three upper bridge arms and the three lower bridge arms comprises a half-bridge circuit or a full-bridge circuit.

20. The solid-state transformer according to claim 19, wherein each of the submodules of each of the three upper bridge arms and the three lower bridge arms has an input port and an output port and further comprises a damping resistor and a bypass switch electrically connected in series between the input port and the output port.

21. The solid-state transformer according to claim 18, wherein each of the three upper bridge arms and the three lower bridge arms further comprises a switch, and in each of the three upper bridge arms and the three lower bridge arms, the inductor is electrically connected to the corresponding phase of the AC power through the switch.

22. The solid-state transformer according to claim 21, wherein the switch is formed by series-connected insulated gate bipolar transistors or series-connected metal-oxide-semiconductor field-effect transistors.

23. The solid-state transformer according to claim 18, wherein each of the three upper bridge arms and the three lower bridge arms further comprises a first switch and a second switch, and the solid-state transformer further comprises two capacitors electrically connected in series across the first DC bus and a midpoint between the two capacitors, wherein in each of the three upper bridge arms and the three lower bridge arms, the inductor is electrically connected to the corresponding phase of the AC power through the first switch, and the inductor is electrically connected to the midpoint through the second switch.

24. The solid-state transformer according to claim 23, wherein each of the first switch and the second switch is formed by series-connected insulated gate bipolar transistors or series-connected metal-oxide-semiconductor field-effect transistors.
